Pathological Gait

5 functional categories that impair a person's ability to walk

1. Deformity — contracture
2. Muscular problems — weakness
   — spasticity
3. Sensory loss
4. Pain
5. Impaired motor control

Deformity – Tissues impair passive mobility
Prevents normal postures & ROM @ the joints for walking
range of motion

Contracture – fibrous connective tissue undergoes structural change because of prolonged inactivity, scarring from injury
Tissues resist passive motion

eg. knee flexion contracture
- inhibits advancement of thigh over tibia in stance
- requires greater muscular effort to stabilize the flexed weight-bearing knee
- shortens the step length due to reduced knee extension in terminal swing

Abnormal joint contours

Congenital disorders such as club foot (Talipes equinovarus)

Muscular problems

Weakness - insufficient strength to meet the demands of walking

Causes - neurological impairments like poliomyelitis, muscular dystrophy etc.

atrophy because of disuse

Spasticity - reduced ROM

Sensory problems

Proprioception

Impaired proprioception - person doesn't know the exact location of their hip, knee or foot & the type of contact with the floor

Walking is slow & cautious

Pain – primary cause of musculoskeletal pain is excessive tension in the tissues

Joint pain

Pain & disuse because of it can lead to deformity and muscle weakness.

Impaired motor control – Brain / Spinal cord lesion results in paralysis

Stroke, Traumatic brain injury, infections, tumours

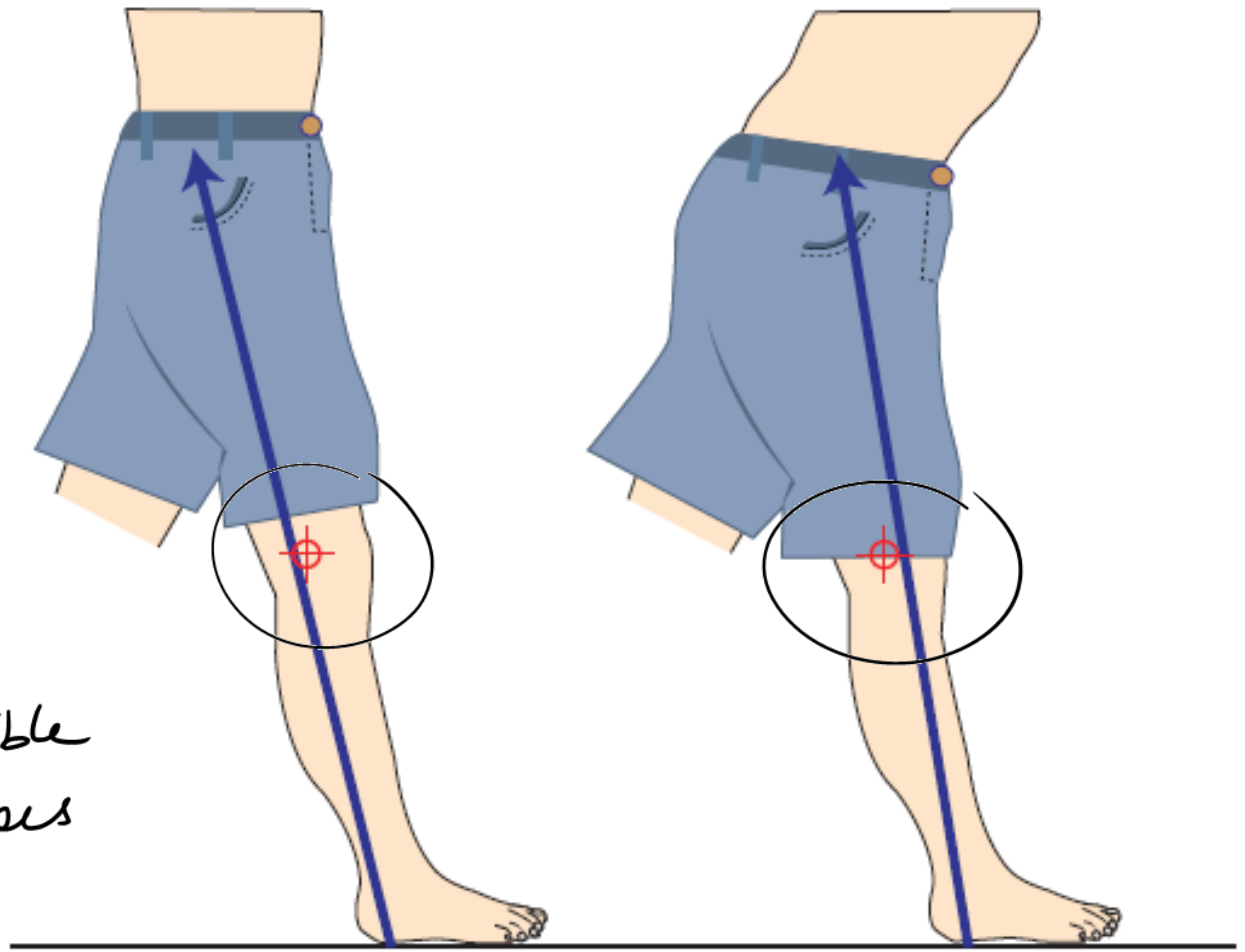
Level of injury determines the level of ability / inability to walk.

Pathological gait – Anterior Trunk Bending

Purpose may be to compensate for weak knee extensors

Person may also place one hand on the thigh, on the affected side to stabilize the knee

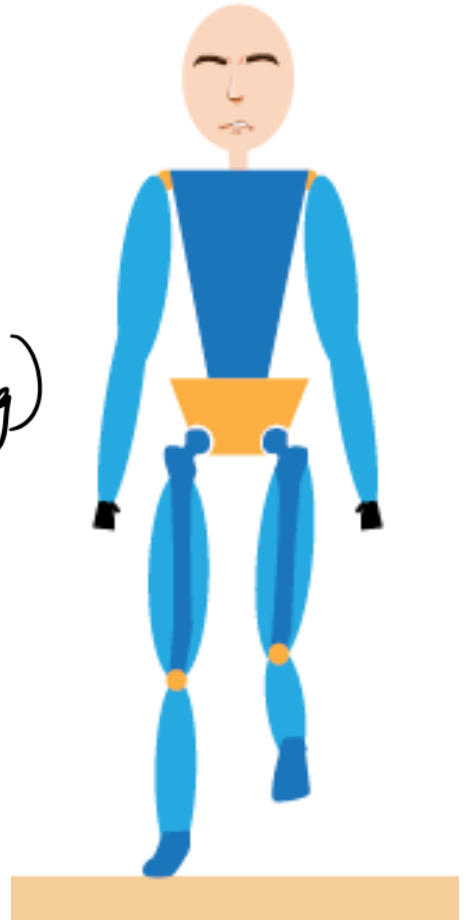
Weak hip extensors
Hip flexion contracture
} possible causes

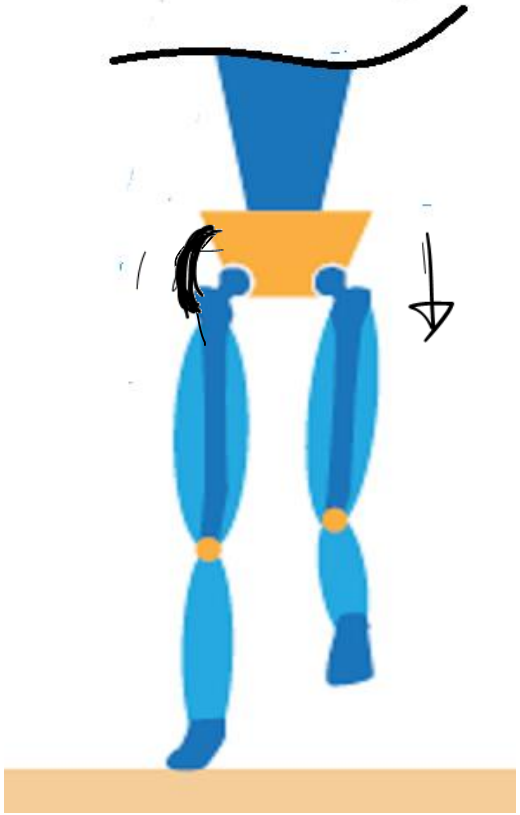


Pathological gait – Lateral Trunk Bending

Best observed in the frontal plane

Unilateral or Bilateral (waddling)





Trendelenburg sign

Hip abductor weakness.

Unable to apply sufficient force to stabilize the pelvis in single stance

In this case, the pelvis will drop on the side which is off the ground

Trendelenburg gait – person uses lateral trunk bending

Ipsilateral lean

Reduces forces in the abductor muscles & hip joint during SS